

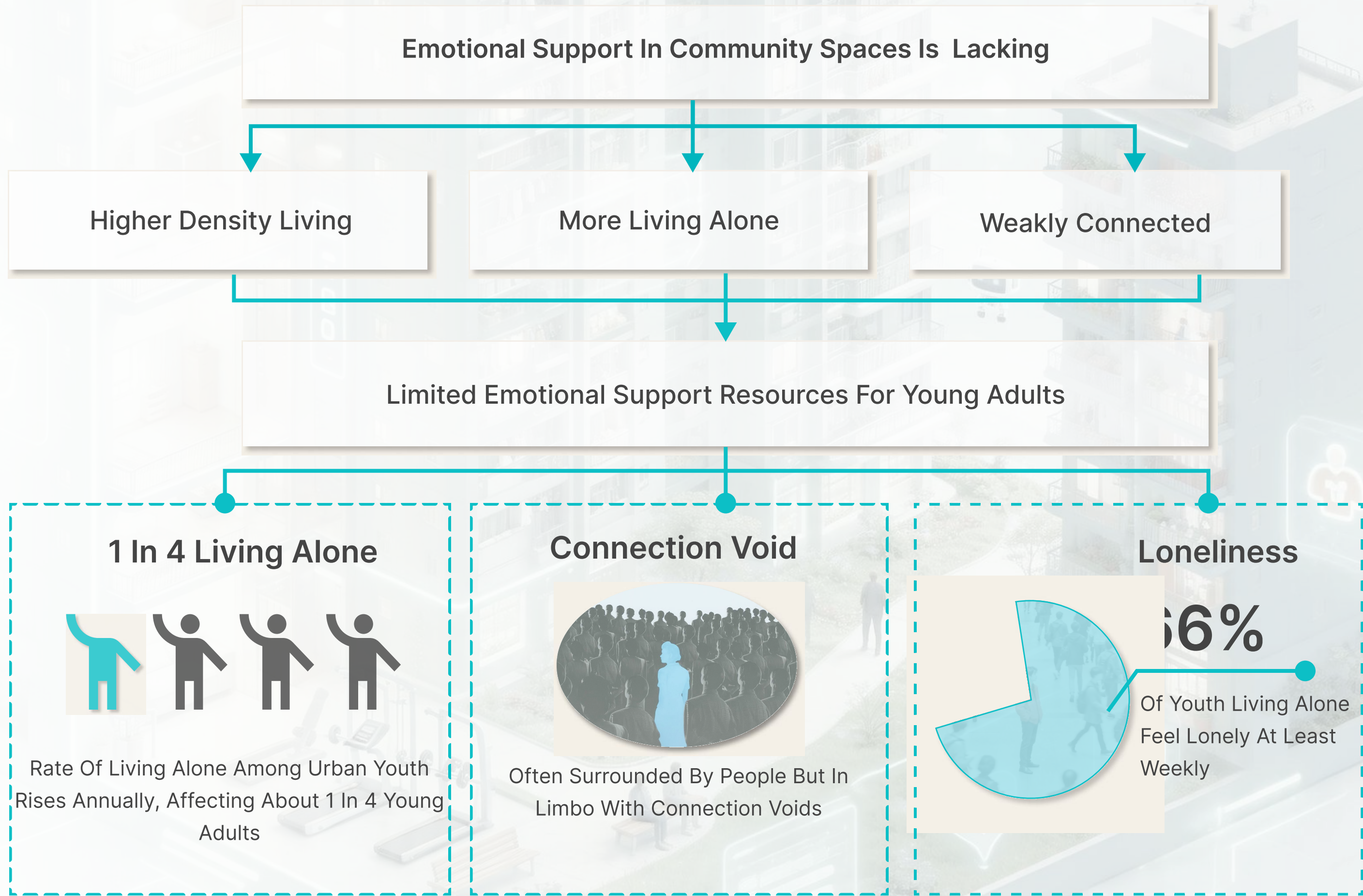
TERRA-LINK

AN URBAN SYMBIOSIS SYSTEM FOR WEAK-TIE ACTIVATION AND COMMUNITY CARE

This project addresses the gap in support experienced by young individuals living alone in high-density cities during moments of “invisible vulnerability,” including emotional fluctuation, disrupted routines, and hesitation to seek help. It proposes a “weak-tie activation and pre-support system.” Through low-threshold access, lightweight connections, and layered support mechanisms, the system provides early reminders, companionship, referrals, and resource matching before crises emerge, enabling timely intervention and enhancing users’ sense of security, connection, and everyday resilience.

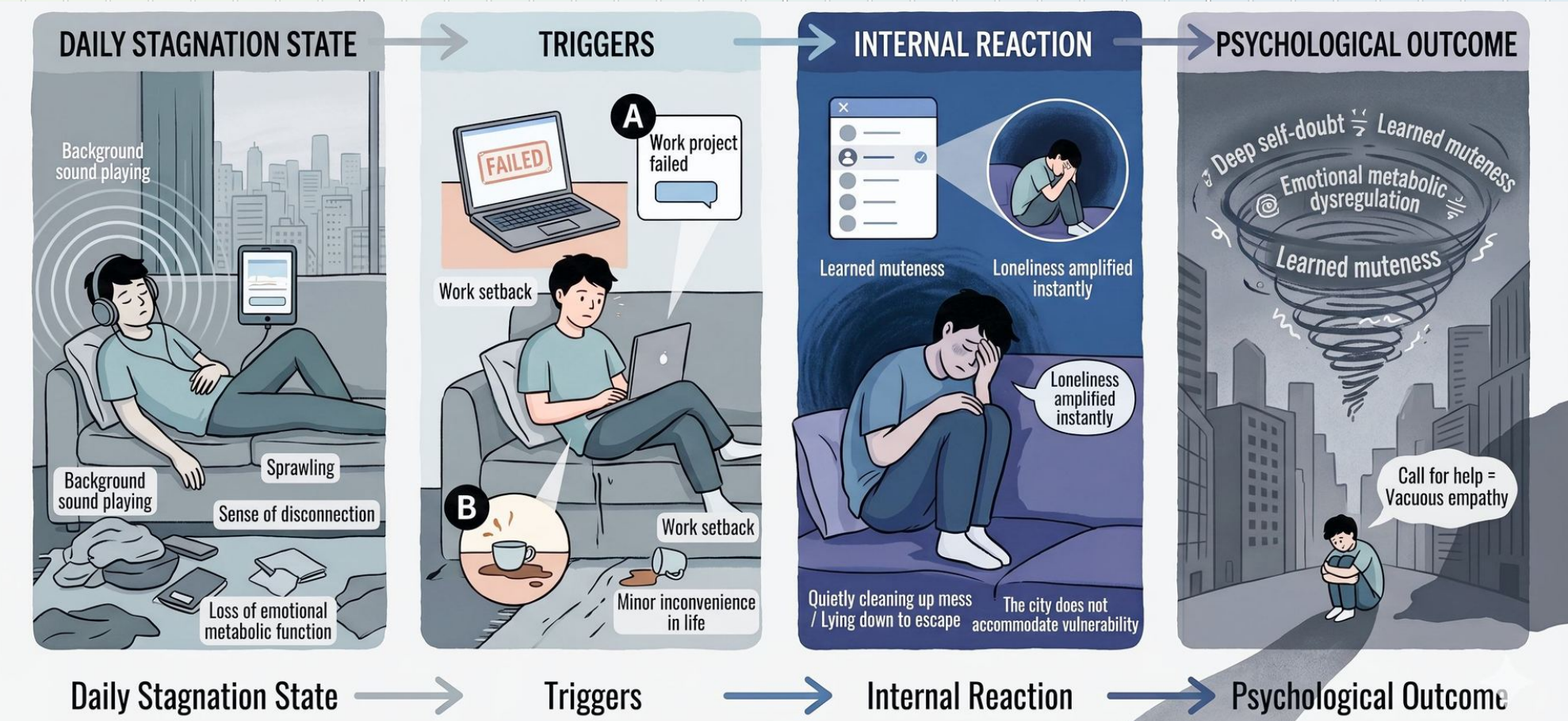
ISSUE BACKGROUND

In High-Density Urban Areas, An Increasing Number Of Young Adults Live Alone Or In Weakly Connected States, Lacking Sustainable Spaces For Daily Emotional Support.



While Pulic Spaces Exist,They Lack Low-Pressure, Non-Intrusive, And Sustainable Mechanisms Foremotional Support.

SCENARIO ANALYSIS



WHERE SUPPORT BREAKS DOWN IN A COMMUNITY REST AREA

- For solitary youth and renters: how low-pressure emotional support fails to emerge in a shared residential space.

- The issue is not the absence of a communal space, but the absence of mechanisms that facilitate emotional support and light social connection.

Breakpoints in Shared Community Rest Area

	 Passing Through	 Pausing Briefly	 Sitting/Observing	 Wanting Connection but Hesitating	 Leaving
Observable Scene	Residents pass by benches and basic facilities daily	Stops for a moment, checks phone	Sits down, watches others in the area	No clear, low-barrier way to engage	Leaves and returns to routine unchanged
Internal State	Tired, wants a brief pause, unsure if it's okay to stop	Wants a break, feels a bit lonely	Desires company, but stays reserved	Wants to say something, but fears rejection	Emotion unchanged, no sense of belonging
Breakpoint	Tired, wants a brief pause, unsure if it's okay to stop	Pause is empty, emotionally unfulfilling	Sees others, but no connection forms	Needs support, but can't reach out	Low-pressure support system absent
Opportunity	Create a subtle reason to pause	Acknowledge brief pauses	Enable subtle social signals	Offer low-pressure optional ways to connect	Build lingering sense of care

Key Gaps in Public Space Support

Space is there, but the “okay to pause” is missing

Simply placing benches isn't enough—there needs to be a low-pressure signal that it's acceptable to pause here

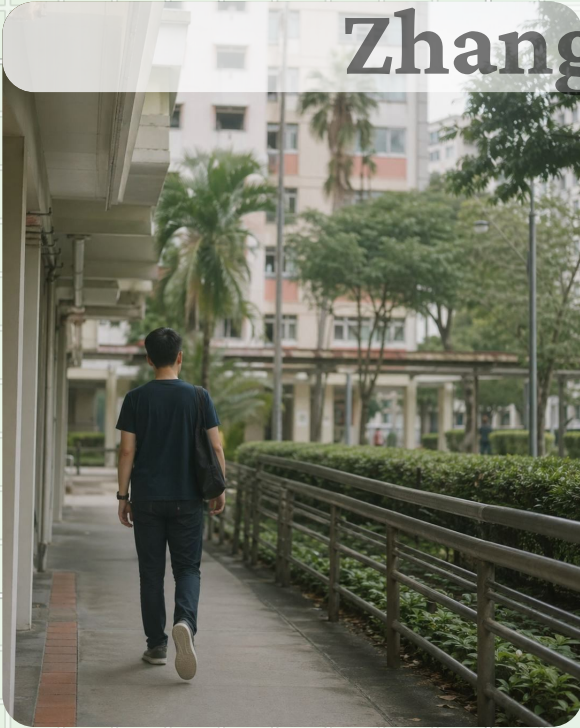
People are present, but mechanisms for light connection are absent

Seeing others isn't enough—there needs to be a way, however small, to feel acknowledged and subtly connected

Space is there, but the “okay to pause” is missing

Emotional needs arise, but without any light, non-intrusive way to seek connection and support, they go unmet

PERSONA



Zhang Yue

Age: 26 | Occupation: Junior financial analyst | City: Singapore
Living: Lives alone in rented HDB room (essentially solitary)
Daily: Fixed 9–6, occasional OT, 45-min commute

Social: Friends in other cities; work-only colleagues; calls family 1–2x/week but hides stress; knows no neighbors

Tech: Heavy smartphone user (short videos, streaming); neutral toward interactive installations



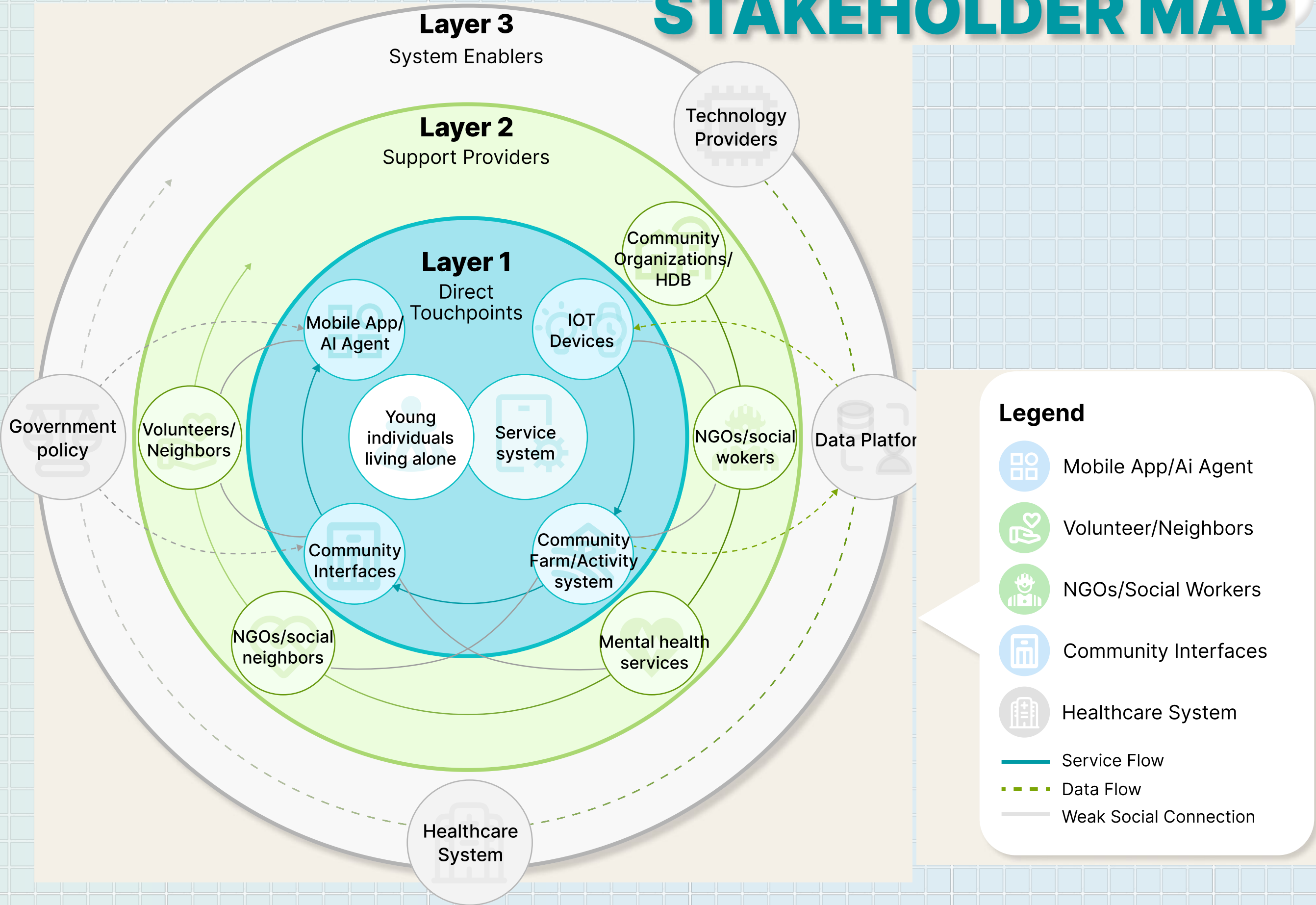
Chen Xiao

Age: 27 | Occupation: Product Operations | City: Shanghai/Shenzhen
Living: Lives alone in high-density youth apartment
Daily: Leaves 9:00, back after 20:00, occasional weekend OT

Social: Moved to SG 1.5 years ago; no stable local circle; family in China; finds meetups exhausting

Tech: Privacy-aware tech user; relies on everyday apps and smart wearables, but rejects intrusive cameras or explicit emotional surveys. Prefers anonymous, opt-in interactions.

STAKEHOLDER MAP



INSIGHT SYNTHESIS MAP



01 SENSING

1. Emotional Changes Are Subtle
2. Lack Of Self-Expression
3. Behavioral Signals Are Weak

Design Direction:

Passive State Sensing



03 INTERVENTION

1. Lack Of Early Intervention
2. Absence Of Everyday Support
3. Delayed Response

Design Direction:

Early, Gentle, Intervention



02 CONNECTION

1. Hesitation To Reach Out
2. Fear Of Bothing Others
3. High Social Barriers

Design Direction:

Low-Pressure Connection



04 ESCALATION

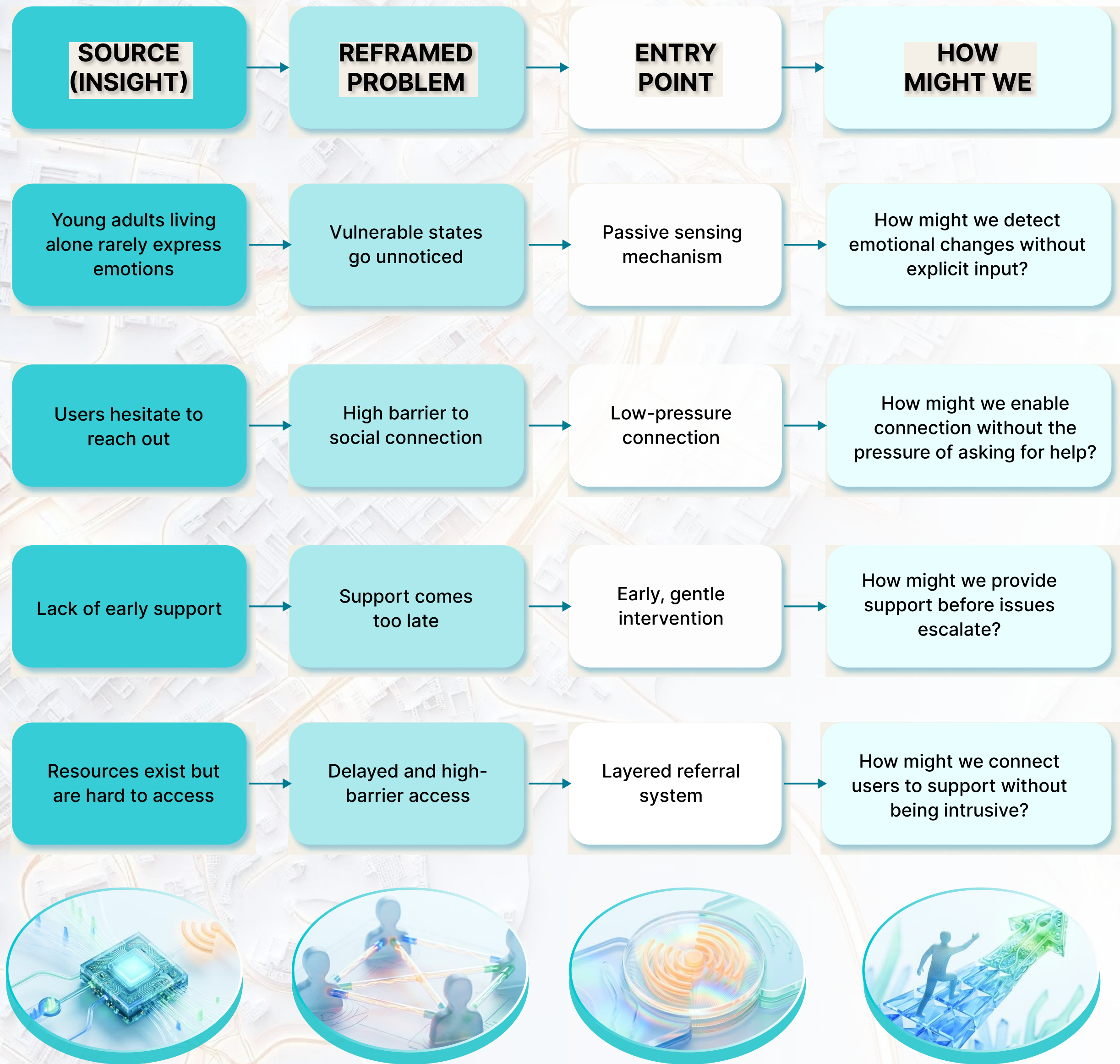
1. Resources Exist But Are Fragmented
2. Unclear Help-Seeking Pathways
3. Difficult Access

Design Direction:

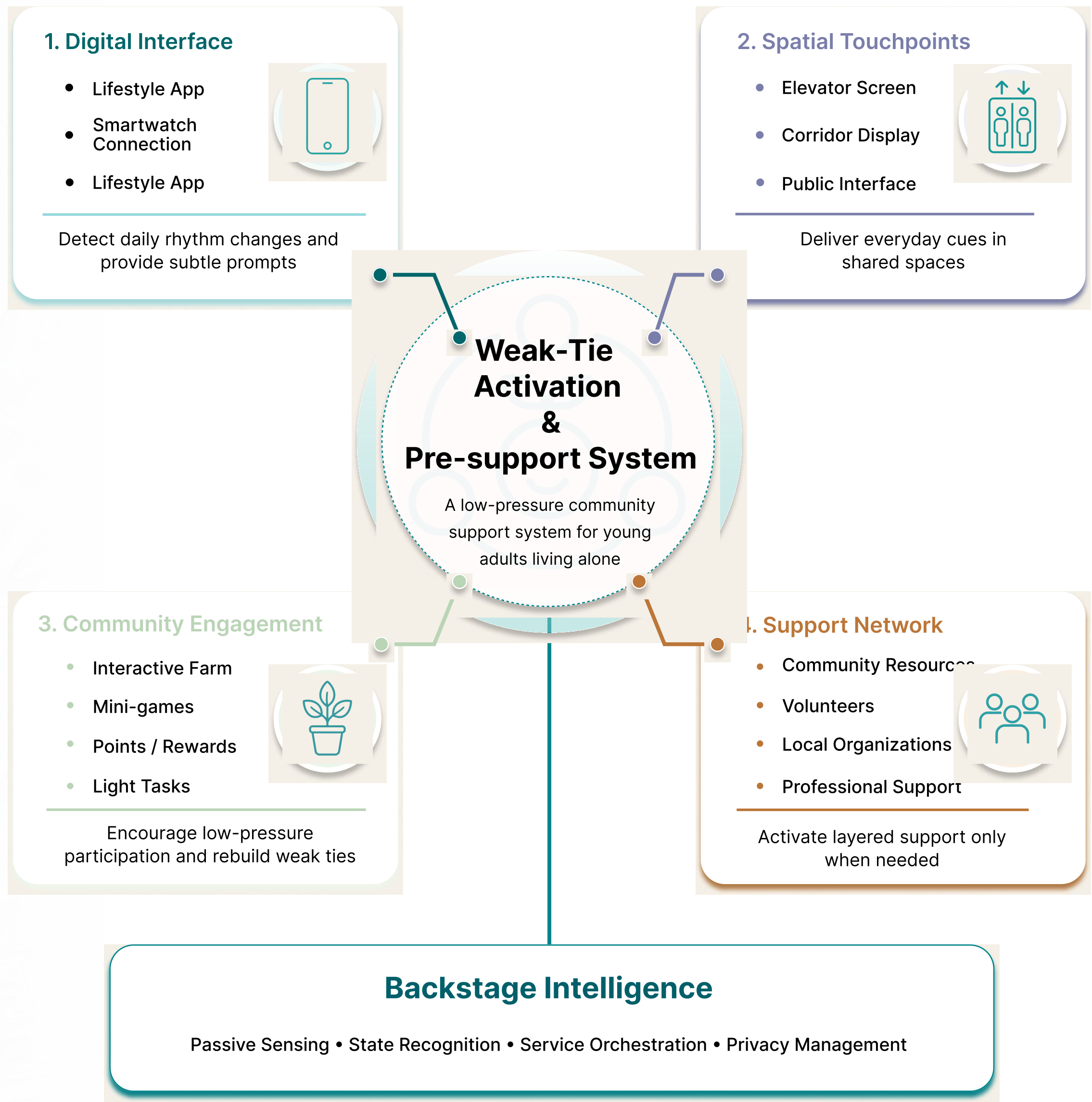
Layered Referral System



DESIGN OPPORTUNITY AREA

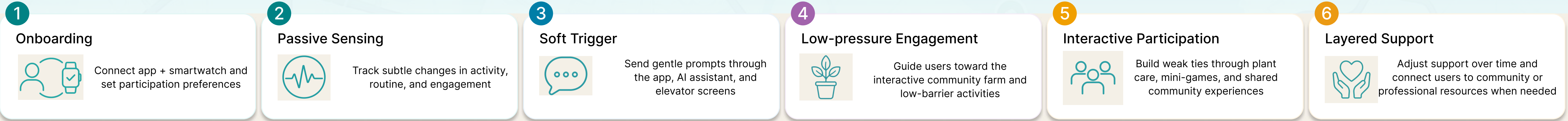


SYSTEM CONCEPT



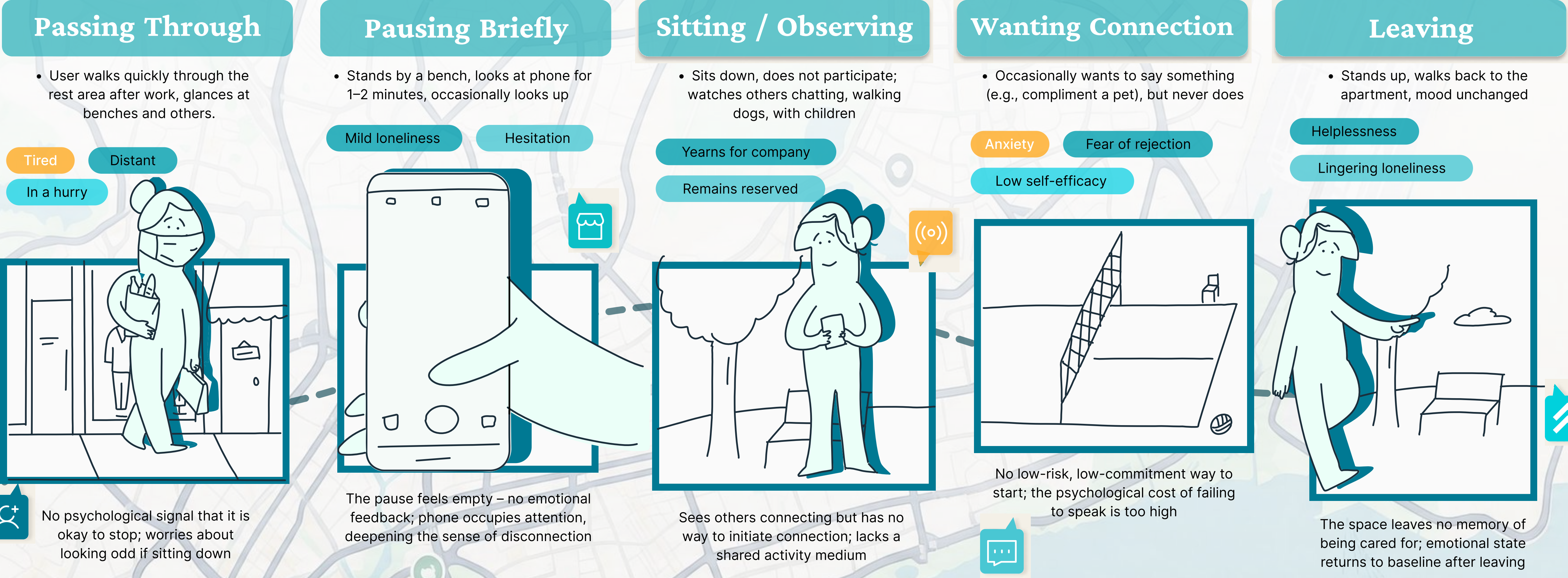
SYSTEM FLOW

Logic : Sense → Nudge → Engage → Connect → Support



USER EXPERIENCE MAP

How can a tired, solitary young adult in a dense community rest area feel: pausing is allowed, loneliness is shared, and connection is low-cost – without force or privacy invasion?



STORYBOARD

From isolation to low pressure community connection



ISOLATION



DIGITAL
PROMPT



SHARED CARE



PARTICIPATION



WEAK TIES



1



Eating Alone

Low energy, low connection



2



Night Reflection

Loneliness, seeking change



3



Wearable Alert

Gentle nudge to connect



4



Activity Notice

Discover nearby activities



5



Elevator Touchpoint

Seamless in daily routine



6



Interactive Check-in

Quick, easy, welcoming



7



Community Farm

Hands-on, side by side



8



AI-supported Participation

Guidance, inclusion, support



9



Exchange & Sharing

Reciprocity, trust, warmth



10



See You Next Time

Light connection, lasting impact

AWARENESS
Recognizing need

MOTIVATION
Recognizing need

ENGAGEMENT
First gentle step

PARTICIPATION
Building connection

BELONGING
Sustained, low-pressure ties

FUTURE DEVELOPMENT STRATEGY

1. From Reactive Support to Predictive Intervention

The system will evolve from passively responding to user needs to proactively identifying signals of “invisible vulnerability” through behavioral patterns (e.g., disrupted routines, reduced mobility, social withdrawal).

2. Humanizing the AI Companion for Long-term Relationships

AI evolves from a functional tool into a companion with contextual memory, building low-pressure trust and guiding users toward human support when needed.

3. Weak-Tie Activation

Enable low-threshold, flexible social interactions through smart matching and context-aware triggers.

4. Spatial Integration

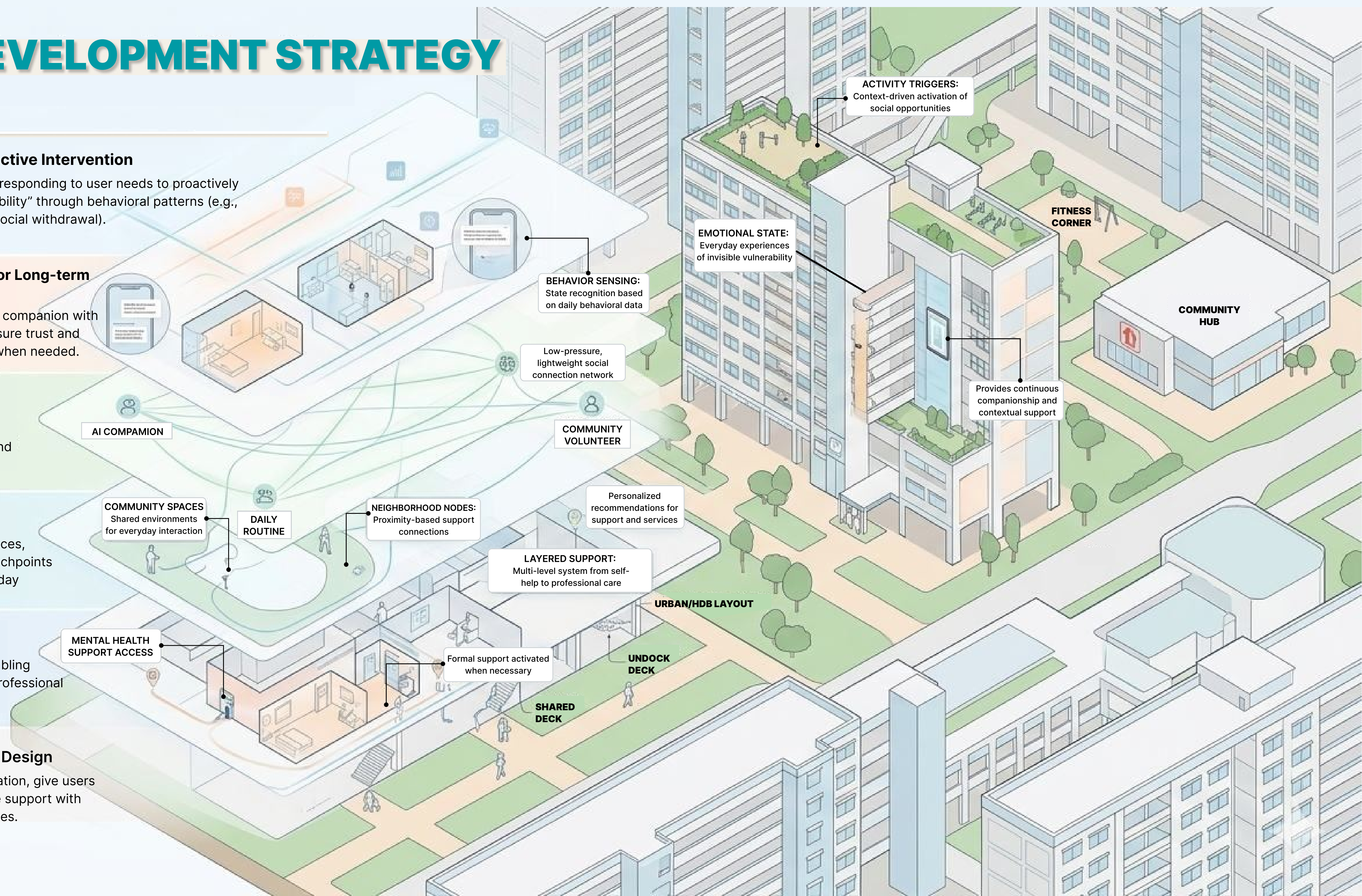
Embed the system into community spaces, linking digital sensing with physical touchpoints to create a seamless, integrated everyday support environment.

5. Layered Support Optimization

Structure a clear support pathway, enabling seamless transition from self-help to professional services.

6. Data Ethics and Low-Intrusion Design

Ensure data transparency and minimization, give users control over interventions, and balance support with privacy by avoiding intrusive experiences.



FINAL VISION

TERRA-LINK

Cultivating Spaces. Strengthening Bonds

Reimagining community life through shared care, nature, and everyday interaction



Awareness

Gentle prompts help residents reconnect with their surroundings and everyday community life.



Participation

Shared farming transforms underused residential spaces into active social infrastructure.



Connection

Repeated care, rewards, and collective activities strengthen neighbourhood bonds.



Terra-Link proposes a future where care is embedded into **everyday living, making community visible, accessible, and participatory.**

